

Vegetable Market (Mandi) Data Analysis

Dataset Overview

- The dataset analyzed contains **1,370 entries** representing **agricultural market (mandi)** data across **19 Indian states**.
- It includes **10 key columns**: State, District, Market, Commodity, Variety, Grade, Arrival_Date, Min_Price, Max_Price, Modal_Price.
- There are **no missing or null values**, indicating clean and well-maintained data suitable for analysis.

Geographical Insights

- **States Represented:** 19
- **Most Active State:** **Kerala** — contributing the highest number of market records (279).
- **Districts Covered:** 121, with **Kangra** being the most represented district (158 entries).
- **Markets Covered:** 178 markets across India, reflecting a broad distribution of mandi-level data.

Commodity Insights

- **Total Commodities:** 119 different commodities are recorded.
- **Most Common Commodity:** **Potato**, with the highest frequency among entries.
- Other widely traded commodities include **Onion, Tomato, Paddy, and Wheat**, indicating a mix of both staple and perishable crops.

Variety and Grade

- **Varieties:** 147 distinct varieties, where “**Other**” dominates with 634 records — likely denoting unspecified or general produce categories.
 - **Grades:** Represent different quality classifications, providing a basis for analyzing price variations by quality.
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Pricing Insights

The dataset records three price measures for each entry:

- **Minimum Price**
- **Maximum Price**
- **Modal Price** (most frequently occurring price)

Findings:

- Prices vary significantly between markets and commodities — suggesting **regional supply-demand differences**.
 - The **modal price** provides a good indicator of standard market value across states.
 - Commodities like **Potato and Onion** show moderate price fluctuations, while **premium produce (like Spices or Pulses)** may exhibit higher variability.
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Temporal Insights

- The Arrival_Date column captures **daily or periodic market arrivals**, enabling **time-series or seasonal trend analysis**.
 - This can help in identifying **price seasonality, harvest cycles, and supply-demand peaks** for key commodities.
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Key Analytical Relationships Identified

- **Commodity vs. Price:** Some commodities (e.g., Potato, Onion) maintain stable modal prices, while others show wide variability.
- **State vs. Commodity:** Regional specialization is visible — e.g., Kerala (spices), Punjab (grains), and Maharashtra (vegetables).

- **District vs. Market Volume:** Markets with higher trade volume tend to have more stable modal prices.
- **Variety vs. Price Range:** Premium varieties or higher grades command higher modal prices.

Conclusion

The analysis of the *Vegetable Mandi (Market)* dataset offers a comprehensive view of India's agricultural trade landscape. It highlights significant **geographical diversity, commodity richness, and market-level price variation** across states.

The data is **well-structured, clean, and suitable** for further analytical modeling — including **price forecasting, seasonal trend analysis, and market behavior prediction**.

Kerala and Kangra emerge as data-intensive regions, while commodities like **Potato and Onion** dominate in trade frequency.

This dataset serves as a **strong foundation for agricultural market analytics**, supporting data-driven decision-making for farmers, traders, and policymakers in improving **price transparency and regional market efficiency**.